

HealthConnect Clinic — Week 7 Project Summary

Data Analytics Track | AnalystLab Africa Experience Lab

1. What Was Planned to Test

Following on from the Week 6 Advanced Analytics & Decision Support Report, Week 7 planned to test: whether the five core KPIs, effect sizes and multivariate model reproduce from raw data; whether the Sunday-anomaly and reminder-channel claims hold up under a formal statistical test rather than a descriptive comparison; whether the high-risk segment definition is robust to threshold choice; and whether the self-built baseline model's Week 6 single-split result is stable under cross-validation.

2. What Was Actually Tested

- Reproducibility of all five KPIs, the Cramér's V effect-size ranking, and the multivariate logistic regression odds ratios, recomputed directly from the raw dataset.
- Formal chi-square significance testing of the Sunday-anomaly and reminder-channel claims (previously supported only descriptively).
- Sensitivity of the highest-priority patient segment to changes in its defining thresholds.
- Stability of the self-built Logistic Regression and Random Forest models under 5-fold stratified cross-validation, compared against the Week 6 single train/test split.
- Error analysis of the Logistic Regression model's false negatives and false positives.
- Stability of feature-importance ranking using permutation importance across cross-validation folds.

3. Most Important Testing Results

- All KPIs, effect sizes and the multivariate model reproduced exactly from raw data — no calculation errors found.
- The reminder-channel claim was only partially supported: SMS significantly beats WhatsApp ($p=0.023$), but not Email ($p=0.448$) — the omnibus three-channel test was not significant ($p=0.098$).
- The Sunday-anomaly conclusion (no KPI distortion) was formally confirmed ($\chi^2=2.01$, $p=0.365$).
- The self-built Logistic Regression model was stable across folds (recall $66.0\%\pm 1.9\%$) and consistently outperformed Random Forest on every fold.
- The high-risk-segment recall estimate, previously based on a too-small $n=20$ test set, was validated on a larger effective sample ($n=66$ via cross-validation): recall 100%, precision 81.8%.

4. Issues or Weaknesses Identified

- The Week 6 reminder-channel recommendation overstated the evidence — it holds specifically for SMS vs WhatsApp, not against Email.
- The 80.9% high-risk-segment no-show rate is sensitive to the exact threshold used and is based on a small sample ($n=68$).
- The Logistic Regression model shows a lead-time-dependent error pattern: it under-flags shorter-lead-time no-shows and over-flags longer-lead-time attendees.

- Data Science still has not responded to the Week 6 outreach — this cross-track dependency remains open.

5. Improvements/Refinements Made

- Narrowed the reminder-channel recommendation to "prioritise SMS over WhatsApp specifically," rather than presenting SMS as unambiguously the best channel.
- Reframed the highest-priority segment as a graduated risk gradient across threshold combinations, rather than a single headline percentage, with the small sample size flagged wherever the 80.9% figure is used.
- Selected Logistic Regression over Random Forest as the primary candidate model, based on its consistently stronger and more stable cross-validated performance.
- Added formal chi-square tests to the Sunday-anomaly and reminder-channel conclusions in place of descriptive comparisons only.

6. Retesting Results

The reminder-channel and high-risk-segment refinements were retested against the same underlying data: the SMS-vs-WhatsApp effect held under a direct pairwise chi-square test, and the graduated segment framing was checked across four threshold combinations rather than one. The Logistic Regression model's stability was retested across all five cross-validation folds individually, confirming the preference over Random Forest was not driven by a single favourable fold.

7. Track(s) Collaborated With

Data Science (continuing the open Week 6 dependency — see Section 9 below for details, since no new two-way exchange occurred this week).

8. What Was Tested Collaboratively

Whether the Week 6 self-built model estimate (from a single train/test split, with an under-powered $n=20$ high-risk-segment test) was reliable enough to serve as a benchmark for Data Science's own model, once they respond.

9. What Changed as a Result

The single-split estimate was replaced with a 5-fold cross-validated benchmark: Logistic Regression recall of 66.0% ($\pm 1.9\%$ across folds), and a high-risk-segment recall of 100% ($n=66$, precision 81.8%) — a validated, larger-sample figure Data Science or ML Engineering can now benchmark against, rather than the smaller, single-split Week 6 numbers.

10. Key Findings / Validation Outcomes

- Every Week 6 KPI, effect size and model coefficient is reproducible from raw data — no errors found.
- Two Week 6 claims needed evidence-based refinement (reminder channel, high-risk segment framing) rather than being taken at face value.
- The self-built model is stable, not a lucky single split, and Logistic Regression is confirmed as the stronger of the two candidates tested.
- A new, previously undetected model-error pattern (lead-time-dependent miscalibration) was identified for Week 8 follow-up.

11. Major Challenges Encountered

The main challenge was testing cross-track dependencies without a live counterpart: with Data Science still not having responded, meaningful cross-track testing had to be constructed as a self-directed continuation of the open dependency (re-testing the model estimate that would eventually be handed off) rather than a live two-way validation exercise.

12. Important Decisions Made

- Chose to run 5-fold cross-validation rather than simply re-reporting the Week 6 single-split numbers, since a single split cannot distinguish a stable result from a lucky one.
- Chose to report the reminder-channel and high-risk-segment findings with their statistical caveats explicitly, rather than preserve the stronger-sounding Week 6 phrasing.
- Chose to flag the model's error pattern as a Week 8 action item rather than attempt a threshold retuning this week, to keep Week 7 focused on testing and diagnosis rather than a new development cycle.

13. Remaining Limitations

- All findings remain correlational, from a single synthetic dataset snapshot.
- The cross-track testing this week is self-directed, not a two-way exchange with Data Science, and should be treated as provisional until they respond.
- The lead-time-dependent model error pattern was identified but not yet corrected.

14. Remaining Issues or Dependencies

- Data Science has still not responded to the Week 6/7 outreach — open dependency carried into Week 8.
- The Sunday-appointment anomaly (737 records) still needs operational/stakeholder clarification.
- Distance/waiting-time missingness (<2% each) still has no agreed imputation approach.

15. Contribution to the Overall HealthConnect Project

This week moves the Data Analytics component from validated to tested: every core finding has either been reproduced from raw data, backed with a formal statistical test, or checked for stability under resampling. Two recommendations were corrected before they could reach final integration in a form the evidence didn't fully support, and the model evaluation now stands on a cross-validated, larger-sample basis that Data Science or ML Engineering can use with confidence.

16. What Must Be Completed Before Week 8

- Close the Data Science loop if a response arrives, and benchmark their model against the cross-validated Logistic Regression figures produced this week.
- Escalate the Sunday-appointment anomaly for a definitive resolution.
- Decide whether to pursue a probability-threshold adjustment for the Logistic Regression model ahead of final integration.
- Carry the refined reminder-channel and high-risk-segment framing into the final HealthConnect recommendation set.